Two ultrametric answers: sharp Grothendieck constant and failure of restricted invertibility

Full answers to Problems 1.4 and 3.2 in arXiv:2503.20972

Automated mathematics discovery run `fa_banach_001`
Lane 5, candidate for human review

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Status: candidate full solution, likely valid. Problem 1.4 holds over every non-Archimedean valued field with the sharp constant $K_{\mathbb{K}} = 1$. Problem 3.2 is false over every such field. This packet does not claim an answer to Problem 1.5 or Problem 2.5.

1 The source problems

Krishna asks three families of questions motivated by classical Euclidean inequalities [1]. The first target here is Problem 1.4: under a normalized scalar bilinear estimate, does the corresponding inner-product estimate hold uniformly over all p-adic Hilbert spaces? The complete source statement is reproduced below.

$$\|(x_j)_{j=1}^d\| := \max_{1\le j\le d} |x_j|, \quad \forall (x_j)_{j=1}^d \in \mathbb{Q}_p^d.$$

Based on Theorem 1.1, we formulate following problem.

Problem 1.4. ***(p-adic Grothendieck Inequality Problem)*** *Let $\mathbb{K}$ be a non-Archimedean valued field. Whether there is a universal constant $K_{\mathbb{K}}$ satisfying the following property: for every p-adic Hilbert space $\mathcal{X}$ over $\mathbb{K}$, for all $m, n \in \mathbb{N}$, if $[a_{j,k}]_{1\le j\le m, 1\le k\le n} \in \mathbb{M}_{m\times n}(\mathbb{K})$ satisfies*

$$\left|\sum_{j=1}^{m}\sum_{k=1}^{n} a_{j,k} s_j t_k\right| \le \left(\max_{1\le j\le m} |s_j|\right)\left(\max_{1\le k\le n} |t_k|\right), \quad \forall s_j, t_k \in \mathbb{K},$$

then

$$\left|\sum_{j=1}^{m}\sum_{k=1}^{n} a_{j,k} \langle u_j, v_k\rangle\right| \le K_{\mathbb{K}}\left(\max_{1\le j\le m} \|u_j\|\right)\left(\max_{1\le k\le n} \|v_k\|\right), \quad \forall u_j, v_k \in \mathcal{X}.$$

The second target is Problem 3.2, the proposed non-Archimedean analogue of the Bourgain–Tzafriri restricted-invertibility theorem.

such that

$$\left\|\sum_{j\in\sigma}a_jTe_j\right\|^2\geq A\sum_{j\in\sigma}|a_j|^2,\quad \forall a_j\in\mathbb{R},\forall j\in\sigma.$$

Recently, Spielman and Srivastava gave a simple proof of Theorem 3.1 by proving a generalization of Theorem 3.1 [18] due to Vershynin [19]. Based on Theorem 3.1, we formulate following problem.

Problem 3.2. ***(p-adic Bourgain-Tzafriri Restricted Invertibility Problem)*** *Let $\mathbb{K}$ be a non-Archimedean valued field. Whether there are universal constants $A>0$, $c>0$ (which may depend upon $\mathbb{K}$) satisfying the following property: If $d\in\mathbb{N}$, and $T:\mathbb{K}^d\to\mathbb{K}^d$ is a linear operator with $\|Te_j\|=1$, $\forall 1\leq j\leq d$, then there exists a subset $\sigma\subseteq\{1,\dots,d\}$ of cardinality*

$$\operatorname{Card}(\sigma)\geq\frac{cd}{\|T\|^2}$$

such that

$$\left\|\sum_{j\in\sigma}a_jTe_j\right\|^2\geq A\max_{j\in\sigma}|a_j|^2,\quad \forall a_j\in\mathbb{K},\forall j\in\sigma.$$

Throughout, $\mathbb{K}^d$ carries the standard maximum norm $\|x\|_\infty=\max_j|x_j|$, as in the source's finite-dimensional model.

2 Results

Theorem 1 (Sharp normalized Grothendieck inequality)**.** *Let $\mathbb{K}$ be any non-Archimedean valued field and let $\mathcal{X}$ be any p-adic Hilbert space over $\mathbb{K}$ in the sense of [1]. Suppose $A=(a_{j,k})\in\mathbb{M}_{m\times n}(\mathbb{K})$ satisfies*

$$\left|\sum_{j=1}^m\sum_{k=1}^n a_{j,k}s_jt_k\right|\leq\left(\max_j|s_j|\right)\left(\max_k|t_k|\right)\qquad(s_j,t_k\in\mathbb{K}).\tag{1}$$

Then, for all $u_j,v_k\in\mathcal{X}$,

$$\left|\sum_{j=1}^m\sum_{k=1}^n a_{j,k}\langle u_j,v_k\rangle\right|\leq\left(\max_j\|u_j\|\right)\left(\max_k\|v_k\|\right).\tag{2}$$

Consequently Problem 1.4 has the optimal answer $K_{\mathbb{K}}=1$ for every $\mathbb{K}$.

Theorem 2 (Failure of p-adic restricted invertibility)**.** *Let $\mathbb{K}$ be any non-Archimedean valued field. There do not exist positive real constants A,c satisfying the assertion in Problem 3.2, even when the constants are allowed to depend on $\mathbb{K}$.*

3 Proof intuition

Both answers have the same cause. In a non-Archimedean norm, a sum is no larger than its largest summand. Thus a normalized scalar Grothendieck hypothesis bounds every matrix entry separately, and no dimension-dependent accumulation remains in the vector-valued sum. The best constant collapses to one.

The same maximum principle makes restricted invertibility fail dramatically. Arbitrarily many identical unit columns define an operator of norm one: their sum still cannot exceed the largest input coordinate. But every restriction containing two columns has a nontrivial kernel. Hence the proposed stable-rank cardinality lower bound and injective lower estimate cannot coexist.

4 Proof of Theorem 1

Let $e_j \in \mathbb{K}^m$ and $e_k \in \mathbb{K}^n$ denote coordinate vectors. Substitute $s = e_j$ and $t = e_k$ in (1). Since zero coordinates are permitted, this gives

$$|a_{j,k}| \leq 1 \qquad (1 \leq j \leq m,\ 1 \leq k \leq n). \tag{3}$$

The strong triangle inequality, (3), and axiom (iv) in the source's definition of a p-adic Hilbert space now give

$$\begin{aligned}\left|\sum_{j,k} a_{j,k}\langle u_j, v_k\rangle\right| &\leq \max_{j,k} |a_{j,k}|\,|\langle u_j, v_k\rangle| \\ &\leq \max_{j,k}\|u_j\|\,\|v_k\| \\ &= \left(\max_j\|u_j\|\right)\left(\max_k\|v_k\|\right).\end{aligned}$$

This proves (2) with $K_{\mathbb{K}} = 1$.

The constant cannot be smaller. Take the one-dimensional p-adic Hilbert space $\mathcal{X} = \mathbb{K}$, put $m = n = 1$, and take $a_{1,1} = u_1 = v_1 = 1$. The scalar hypothesis holds and both sides of (2) equal one. This proves sharpness.

Remark 3. *The argument also shows that the scalar hypothesis (1) is equivalent to the entrywise condition* $\max_{j,k} |a_{j,k}| \leq 1$*. The reverse implication follows immediately from the strong triangle inequality. This is precisely where the normalized non-Archimedean problem differs from its Archimedean analogue.*

5 Proof of Theorem 2

For each $d \in \mathbb{N}$, define $T_d : \mathbb{K}^d \to \mathbb{K}^d$ by

$$T_d(x_1, \ldots, x_d) = \left(\sum_{j=1}^{d} x_j\right) e_1. \tag{4}$$

Every column is the same unit vector:

$$T_d e_j = e_1, \qquad \|T_d e_j\|_\infty = 1. \tag{5}$$

Moreover, ultrametricity gives

$$\|T_d x\|_\infty = \left|\sum_{j=1}^{d} x_j\right| \leq \max_j |x_j| = \|x\|_\infty.$$

Thus $\|T_d\| \leq 1$, while equality holds at $x = e_1$; hence

$$\|T_d\| = 1. \tag{6}$$

Suppose for contradiction that Problem 3.2 held with some fixed $A, c > 0$. Choose an integer d such that $cd \geq 2$ and apply it to T_d. By (6), the promised set σ would satisfy

$$\mathrm{Card}(\sigma) \geq \frac{cd}{\|T_d\|^2} = cd \geq 2.$$

Choose distinct $j, k \in \sigma$, set $a_j = 1$, $a_k = -1$, and set all other a_i to zero. From (5),

$$\sum_{i\in\sigma} a_i T_d e_i = e_1 - e_1 = 0, \qquad \max_{i\in\sigma} |a_i|^2 = 1.$$

The demanded estimate would therefore read $0 \geq A$, contradicting $A > 0$. This proves Theorem 2.

The cancellation is valid in every characteristic. In characteristic two, $-1 = 1$, but $1 + (-1) = 0$ and $|-1| = 1$ still hold.

6 Scope and novelty check

Theorem 1 answers Problem 1.4 only. Problem 1.5 constrains every scalar test coordinate to have modulus one, so the coordinate probes with zeros used in (3) are unavailable; no answer to that formulation is claimed here.

Problem 2.5 prints the same factor $1 - \varepsilon$ in its lower and upper distortion bounds, apparently a typographical error. We neither correct nor exploit that display. Theorem 2 answers Problem 3.2 exactly as printed.

The run's cheap indexes were searched by arXiv identifier, exact title, and the core problem phrases. A bounded primary arXiv search found the source paper but no later resolution of either problem. This is not an exhaustive novelty certification.

References

[1] K. Mahesh Krishna, *p-adic Grothendieck Inequality, p-adic Johnson–Lindenstrauss Flattening and p-adic Bourgain–Tzafriri Restricted Invertibility Problems*, arXiv:2503.20972v3 (2026).